

THE VIBRATION OF FRAMES

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In this paper a method is presented for finding the natural frequencies and principal modes of undamped free vibration of a plane frame composed of a rectangular grid of uniform beams. A general method is presented for finding a set of assumed modes which may be used in a Rayleigh-Ritz analysis of the system. It is shown how the inertia and stability matrices for these modes may be found, and how the whole analysis may be set up in a matrix form, suitable for use with a digital computer. In the final section the method is applied to a simple frame and the results obtained are compared with the exact solution.

1. INTRODUCTION

During a recent investigation into the vibration and noise transmission characteristics of reinforced concrete building structures it was found to be desirable to determine analytically a number of the natural frequencies and normal modes of undamped free vibration of a frame. A survey of the literature showed that, although a variety of techniques had been put forward in recent years for finding natural frequencies of such systems, almost all, if not all, had serious disadvantages. The method discussed in this paper was devised as an attempt to avoid at least some of these disadvantages. The paper should be regarded as a sequel to an earlier one (1). Throughout the paper only the in-plane vibrations of plane frames of rectangular type will be considered. However, all the methods that will be described may be extended to more general plane or space frames. In the following section some of the more generally used methods will be described.

2. A DISCUSSION OF METHODS

It will help if attention is focused on a typical example of a frame. Such an example is the plane, single bay, frame with n storeys shown in Figure 1.

If only the fundamental mode of vibration is needed, and this only to a rough approximation, then one way to find it is to make a lumped mass (or lumped elasticity) model of the structure. Many different models can be used, depending on what kind of model is used to replace each beam. Various possible models of vibrating beams have been discussed by Duncan (2), Livesley (3), Gladwell (4), Leckie and Lindberg (5) and Lindberg (6). The natural frequencies of the model may be found by setting up the appropriate Lagrangian equations which will yield the (squares of the) natural frequencies of the model as the eigenvalues of a matrix equation. Although discrete models may be suitable for finding the fundamental frequency of a frame they are not in general suitable for finding the higher frequencies accurately. This is because so many elements have to be used to replace each continuous member that the equations can easily become unwieldy, even by computer standards. If a discrete method is the only possible one, if for instance the frame is made up of non-uniform beams, then the process of finding the natural frequencies can be made more economical by using the Branch Mode Method discussed

in a previous paper (1). If the higher modes of vibration are required it is essential to take into account the fact that the mass and elasticity of the frame are both distributed continuously. This is borne out by the work of Leckie and Lindberg (5, 6). It would be interesting to see their work on single non-uniform beams extended to frames made up

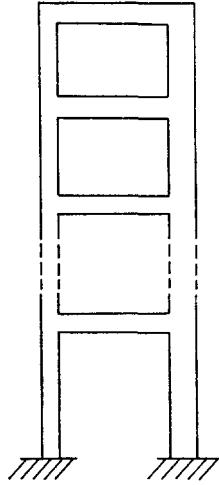


Figure 1. A plane, single-bay, n -storey frame.

of non-uniform beams. In this paper we are concerned primarily with frames made up of uniform beams, and for such there are a number of other possible methods of analysis. First we shall discuss the receptance method described by Bishop (7, 8) and Bishop and Johnson (9).

We shall apply the receptance method to the vibration of a two-storey frame which is symmetrical about a vertical line through the centres of the spans, as shown in Figure 2(a). We shall make two assumptions: first, that members meeting at a joint remain at right

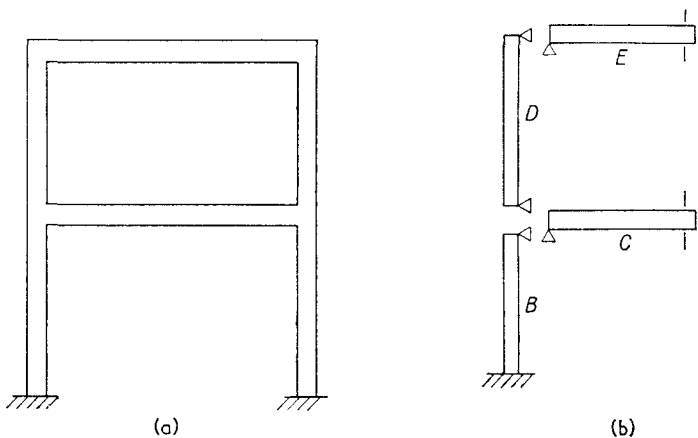


Figure 2. Symmetrical two-storey frame and a simplified system for symmetrical modes.

angles to each other, and secondly, that all extensional vibrations of members may be neglected. The second assumption is justified for all but the very high frequency range; it is retained for simplicity, but can be dropped without introducing too much extra

difficulty. The modes of such a frame are either symmetrical or anti-symmetrical; we shall first consider the symmetrical modes. It may easily be shown that in these modes the mid-points of the cross-members behave as "sliding" ends, and the joints have no horizontal motion, in other words, the beams are "pinned" at the joints. Thus for symmetrical modes it is sufficient to consider the system shown in Figure 2(b). To analyse this system we denote the displacement, slope, shearing force and bending moment at the lower (or left hand) end of a member by q_1 , q_2 , Q_1 and Q_2 respectively, and those at the upper (or right hand) end by q_3 , q_4 , Q_3 and Q_4 . Let β_r , be the receptance linking the q_r and Q_r of the typical member B . Systems B , C and D are all linked together by a single co-ordinate, the slope of the members at the junction, so that the compatibility condition is

$$q_{4b} = q_{2c} = q_{2d} = q_2.$$

Similarly, the compatibility condition at the function between D and E is

$$q_{4d} = q_{2e} = q_4.$$

Because of the symmetry of the system the only equilibrium conditions which need to be considered are those for the bending moments at the junctions, namely

$$Q_{4b} + Q_{2c} + Q_{2d} = 0 = Q_{4d} + Q_{2e}. \quad (2.1)$$

Combining these compatibility and equilibrium equations with the receptance equations

$$\left. \begin{aligned} q_{4b} &= \beta_{44} Q_{4b}, & q_{2c} &= \gamma_{22} Q_{2c}, \\ q_{2d} &= \delta_{22} Q_{2d} + \delta_{24} Q_{4d}, & q_{4d} &= \delta_{42} Q_{2d} + \delta_{44} Q_{4d}, \\ q_{2e} &= \epsilon_{22} Q_{2e}, \end{aligned} \right\} \quad (2.2)$$

we obtain, in the usual way, the three equations

$$\left. \begin{aligned} (\beta_{44} + \gamma_{22}) Q_{4b} + \gamma_{22} Q_{2d} &= 0, \\ \gamma_{22} Q_{4b} + (\gamma_{22} + \delta_{22}) Q_{2d} + \delta_{24} Q_{4d} &= 0, \\ \delta_{42} Q_{2d} + (\delta_{44} + \epsilon_{44}) Q_{4d} &= 0. \end{aligned} \right\} \quad (2.3)$$

The frequency equation is the condition that these three equations should have a non-zero solution, namely

$$\begin{vmatrix} \beta_{44} + \gamma_{22} & \gamma_{22} & \\ \gamma_{22} & \gamma_{22} + \delta_{22} & \delta_{24} \\ & \delta_{42} & \delta_{44} + \epsilon_{22} \end{vmatrix} = 0.$$

The receptances of uniform beams have been presented by Bishop (7). For a frame in which B , C , D and E each have length l (so that the cross members in the whole frame have length $2l$), cross-sectional area A , density ρ , Young's modulus E , and second moment of area of cross-section about the neutral axis through its centroid I , the relevant receptances are as follows:

$$\begin{aligned} \beta_{44} &= F_3/EI\lambda F_5, & \gamma_{22} &= \epsilon_{22} = F_6/2EI\lambda F_2, \\ \delta_{22} &= \delta_{44} = -F_5/2EI\lambda F_1, & \delta_{24} &= F_8/2EI\lambda F_1, \end{aligned}$$

where

$$\lambda = (A\rho\omega^2/EI)^{1/4}$$

and $F_1 = \sin \lambda l \sinh \lambda l$, $F_2 = \cos \lambda l \cosh \lambda l$, etc., as in (7). It is clear that the receptance method may be used to analyse the vibration of any frame. For the symmetrical vibration of a symmetrical n -storey, single bay frame of the type shown in Figure 1 the frequency determinant will have order $2n - 1$.

Very often a frequency equation of lower order may be obtained by expressing the forces and couples at the junctions in terms of the deflections and slopes, instead of vice versa as in the receptance analysis. If this is done for the system shown in Figure 2(a) and equations (2.2) are solved for the Q 's, the result is

$$\begin{aligned} Q_{4b} &= B_{44}q_2, & Q_{2c} &= \Gamma_{22}q_2, & Q_{2d} &= \Delta_{22}q_2 + \Delta_{24}q_4, \\ Q_{4d} &= \Delta_{42}q_2 + \Delta_{44}q_4, & Q_{2e} &= E_{22}q_4. \end{aligned}$$

Substituting these into the equilibrium relations (2.1) we obtain just the two equations

$$\left. \begin{aligned} (B_{44} + \Gamma_{22} + \Delta_{22})q_2 + \Delta_{24}q_4 &= 0, \\ \Delta_{42}q_2 + (\Delta_{44} + E_{22})q_4 &= 0. \end{aligned} \right\} \quad (2.4)$$

The quantities B_{44} , Γ_{22} etc. are called *impedances*. Unfortunately this term has often been used rather vaguely, as may be seen by reading the discussion following the paper by Bishop (7). In section 5 we show how the appropriate impedances may be obtained from Table 1. For the particular system considered above the impedances are

$$\begin{aligned} B_{44} &= \frac{E\lambda F_5}{F_3}, & \Gamma_{22} &= \frac{2E\lambda F_2}{F_6} = E_{22}, \\ \Delta_{22} &= \frac{E\lambda F_5}{F_3} = \Delta_{44}, & \Delta_{24} &= \frac{E\lambda F_8}{F_3}. \end{aligned}$$

The frequency equation for the symmetrical modes of the symmetrical n -storey single bay frame will have order n . It is again clear that impedances can be used to analyse the vibration of any frame.

Another method which must be considered is the transfer matrix method described, for instance, by Pestel and Leckie (10). This may easily be applied to a frame of the type shown in Figure 1, but it is more efficient than the previous methods only when applied to frames of this type, that is, frames which have a predominantly chain-like topology. Even though it could be used, it would have no advantage over the receptance and impedance methods when applied, for instance, to a frame with m bays and n storeys, or to a frame in three dimensions. For this reason we shall not consider the method further.

Although the receptance and impedance methods may be used to analyse any frame made up of uniform beams, they suffer from one great practical disadvantage. The natural frequencies are found as the zeros of a determinant each of whose elements is a transcendental function of the frequency parameter (or parameters) λ . The zeros of such a determinant can be found only by trial and error. Because of the very arbitrary nature of the determinant treated as a function of λ , or ω , only a very crude iteration process, in which the determinant is evaluated at a sequence of points, may be used, and this has been found to take a disproportionate amount of computer time. Moreover, such an iteration process never yields an exact zero of the determinant, so that, at the approximate zero, the frequency equations will not have a solution. This situation occurs in the ordinary eigenvalue problem for a symmetric matrix $\mathbf{A}$ whose elements are independent of ω . There it is found that if the matrix is of only moderate size, say $n \geq 8$, it is not, in general, possible to find the eigenvector corresponding to an approximate eigenvector λ merely by solving $n - 1$ of the scalar equations in the matrix equation

$$(\mathbf{A} - \lambda \mathbf{I}) \mathbf{x} = \mathbf{0}. \quad (2.5)$$

This is because, when λ is an approximate eigenvalue, the determinant of coefficients is very small and the equations are ill-conditioned.

Instead, the eigenvector must be found by using an iterative process which starts from an arbitrary vector $\mathbf{x}_0$ and obtains $\mathbf{x}_r$ from the equations

$$(\mathbf{A} - \lambda \mathbf{I}) \mathbf{x}_r = \mathbf{x}_{r-1}, \quad (r = 1, 2, \dots).$$

It may easily be shown that, under certain conditions, the vectors $\mathbf{x}_r$ converge to the eigenvector associated with the eigenvalue to which λ is an approximation. The whole matter is discussed by Bishop, Gladwell and Michaelson (11). For this ordinary eigenvalue problem there are now highly developed computer programmes which can find the eigenvalues and eigenvectors of matrices of very high order, say $n \sim 100$.

If receptance or impedance analysis is used, the frequency equation will have the form

$$|\mathbf{B}(\omega)| = 0. \quad (2.6)$$

If an approximate mode shape is to be found then some method, similar to that described above, will have to be found for solving the equations

$$\mathbf{B}(\omega) \mathbf{x} = \mathbf{0},$$

which again may be expected to be ill-conditioned. It was because such a method had not been developed, at least to a very great extent, that the author attempted to find a method which would lead to equations of the form (2.5), rather than ones of the form (2.6).†

Such a method is the Rayleigh-Ritz method. In this, as is well known, it is assumed that the system can vibrate only in combinations of a certain set of assumed modes. The practical difficulty in applying the Rayleigh-Ritz method to a complicated system is that of finding a set of modes which gives sufficient freedom to the system, that is, a set which is sufficiently complete. The Branch Mode Method which will now be described provides a systematic means for obtaining a reasonably complete set of modes. The essence of the procedure is that the problem is divided up into a number of stages, as follows.

1. Impose a sequence of sets of constraints to the system in turn. These sets of constraints may conveniently be chosen so that just a few adjoining components vibrate with distortion, the others either remaining fixed or vibrating as rigid bodies, but in either case remaining attached to the distorting components. The various constrained systems which are obtained in this way are called *branches* of the original system.
2. Determine a certain number of the lower natural frequencies and normal modes of each of these branches.
3. Use these modes, which are called *branch modes* (together with the appropriate rigid-body modes if the system is free) in a Rayleigh-Ritz analysis of the whole system.

Two points should be noted. First, the procedure is exactly the same as described in an earlier paper (1), except that for the frames made up of uniform beams which are now being discussed, it is unnecessary to first make a lumped mass model of the system. Secondly, since the modes are found by constraining the system, they will all automatically satisfy the conditions of compatibility which must hold between the displacements and rotations at the joints of the frame. This has the further consequence that it will be known beforehand that the values obtained for the natural frequencies in the final stage of the analysis will all be too high. This result may be proved using the analysis described by Gould (12).

† The transfer matrix method does not suffer from this disadvantage. When the method may be used, the mode shapes are automatically found in the process of the solution.

3. THE CHOICE OF THE BRANCHES

To show how the branches may be chosen it is convenient to refer to a particular system. We shall choose that shown in Figure 2(a). For the symmetrical vibrations it is sufficient to consider the system in Figure 3(a), and for that the branches could be chosen as in

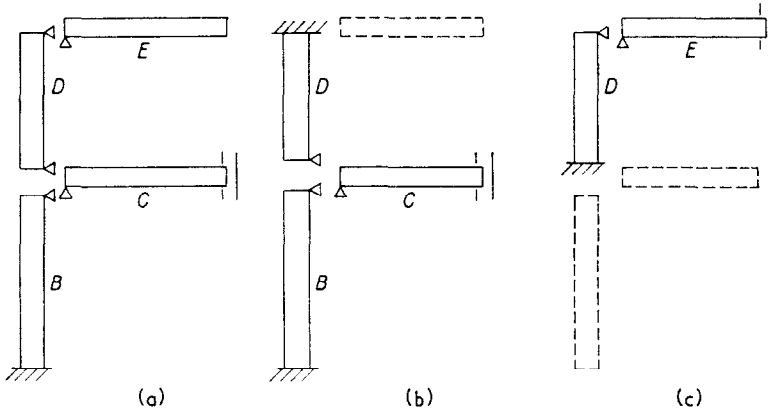


Figure 3. The half-frame for symmetrical modes, and a possible division into branches.

Figures 3(b) and 3(c), where the parts of the system shown in broken lines are assumed to be rigidly fixed. It will be noticed that the branches *overlap*, in the sense that *D* vibrates in each branch. Also branch *b* allows the lower joint to rotate while branch *c* allows the upper one to do so. It may easily be shown that the aggregate of modes obtained from these two branches is complete (see section 7).

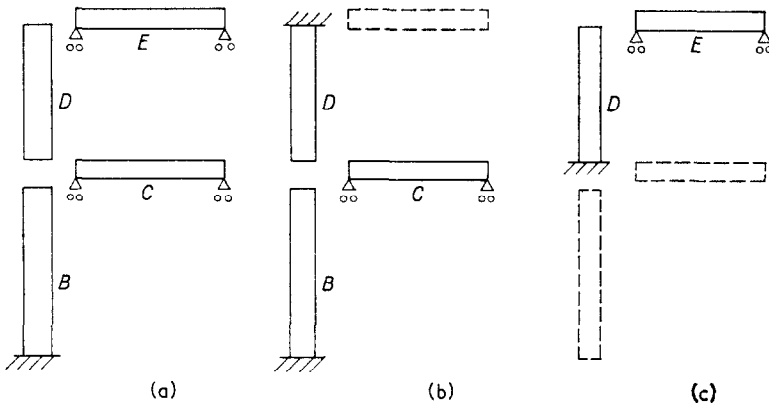


Figure 4. The half-frame for anti-symmetrical modes, and a possible division into branches.

When the frame executes anti-symmetrical vibration, the mid-points of the cross members behave as if they were pinned, the pins being able to move with translational motion in the horizontal direction. It is thus sufficient to consider the half-frame shown in Figure 4(a). A possible set of branches for this system is shown in Figures 4(b) and 4(c). In these branches one end of the upright members is "free", that is, able to deflect transversely as well as rotate. Each of these branches may be replaced by two simpler branches in which these ends are free *either* to deflect without rotating, *or* to rotate without

deflecting. Thus the system may be analysed using the four branches shown in Figure 5. It will be seen that, taken all together, these branches allow each joint to move with its full freedom.

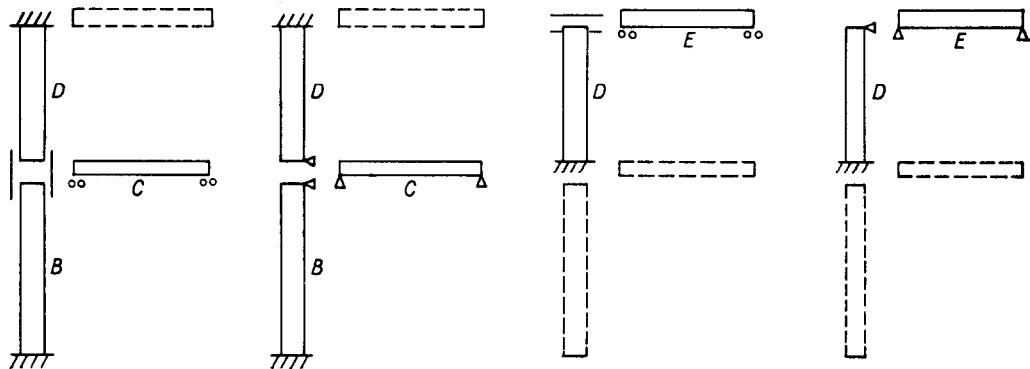


Figure 5. A simpler way to divide the system of Figure 4(a) into branches.

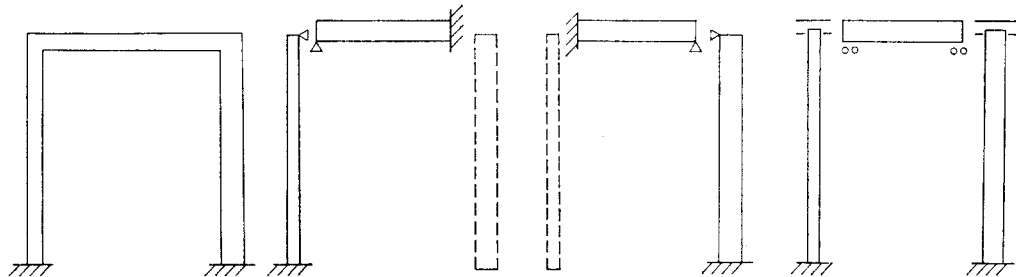


Figure 6. An asymmetrical frame and its division into branches.

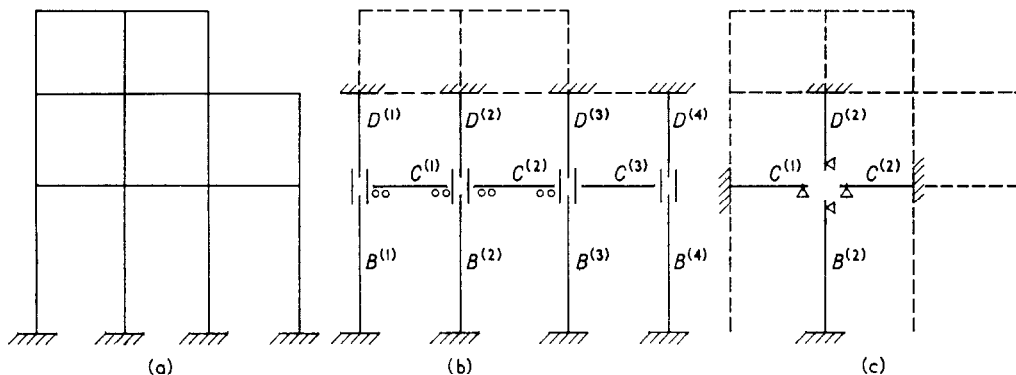


Figure 7. A more complicated system and two typical branches for it.

We shall conclude this section by showing how some other types of frames may be analysed into branches. Figure 6 shows a simple asymmetrical frame, and the three branches (of two types) into which it may be analysed. Figure 7(a) shows a general plane frame. This again has two types of branches. In a branch of the first kind one “floor” is allowed to move horizontally and all the uprights leading to it are restrained from rotating

where they meet that floor and are clamped at the adjacent floors, as in Figure 7(b). In a branch of the second kind one joint is allowed to rotate and the members leading to it are clamped at the adjacent joints, as in Figures 7(c). An indication of the movement which is

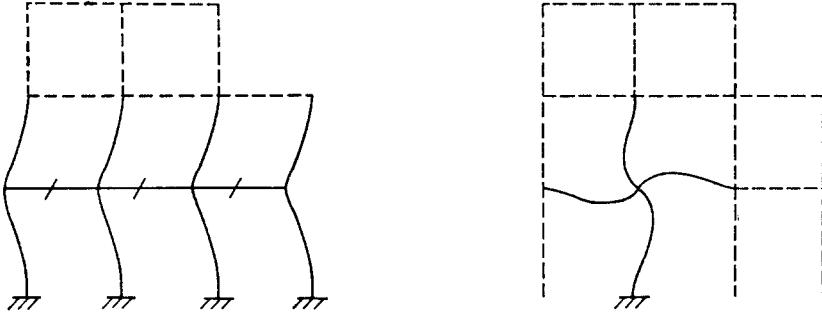


Figure 8. Typical modes for the branches shown in Figure 6.

possible in these branches is shown in Figure 8. It should be remembered however that a number of principal modes of each branch can be used in the final Rayleigh-Ritz analysis so that these diagrams show only a small part of the freedom which the system possesses.

4. THE DETERMINATION OF THE BRANCH MODES

The branches have deliberately been chosen so that they have particularly simple frequency equations when analysed using receptances or impedances.

Consider the branches shown in Figure 3. Using the same notation as in section 2 we find that the receptance equations of the first branch are

$$q_{4b} = \beta_{44} Q_{4b}, \quad q_{2c} = \gamma_{22} Q_{2c}, \quad q_{2d} = \delta_{22} Q_{2d},$$

where now δ_{22} is a receptance of the pinned-clamped beam D , not of a pinned-pinned beam as in equation (2.2). Substituting these equations into the equilibrium equation, the first of equations (2.1), we find

$$\left(\frac{1}{\beta_{44}} + \frac{1}{\gamma_{22}} + \frac{1}{\delta_{22}} \right) q_2 = 0,$$

so that the frequency equation is simply

$$\frac{1}{\beta_{44}} + \frac{1}{\gamma_{22}} + \frac{1}{\delta_{22}} = 0. \quad (4.1)$$

The frequency equation of the second branch is similarly

$$\frac{1}{\delta_{44}} + \frac{1}{\epsilon_{22}} = 0, \quad (4.2)$$

where now δ_{44} is a receptance of the clamped-pinned beam D .

The reader may verify that for the branches shown in Figure 5, the frequency equations are respectively

$$\frac{1}{\beta_{33}} + \frac{1}{\gamma_{55}} + \frac{1}{\delta_{11}} = 0; \quad \frac{1}{\beta_{44}} + \frac{1}{\gamma_{22}} + \frac{1}{\delta_{22}} = 0; \quad \frac{1}{\delta_{33}} + \frac{1}{\epsilon_{55}} = 0; \quad \frac{1}{\delta_{44}} + \frac{1}{\epsilon_{22}} = 0.$$

Here receptances with suffices 55 refer to rigid-body motion of the cross-member. If all the members have the same values of $A\rho$, EI and l the receptances relevant to the first branch of Figure 5 are

$$\beta_{33} = \delta_{11} = -F_3/EI\lambda^3 F_6, \quad \gamma_{55} = -1/A\rho l\omega^2.$$

Now consider the frequency equations for the branches shown in Figure 7. For the first branch let q_1 denote the translational displacement of the floor, and consider the equilibrium equation for the total horizontal force on the floor. The cross members will provide a total force

$$Q_{5c} = \sum_{r=1}^3 (1/\gamma_{55}^{(r)}) q_1,$$

while the uprights below and above respectively will provide forces

$$Q_{3b} = \sum_{r=1}^4 (1/\beta_{33}^{(r)}) q_1, \quad Q_{1d} = \sum_{r=1}^4 (1/\delta_{11}^{(r)}) q_1.$$

Thus the frequency equation is

$$\sum_{r=1}^4 \left(\frac{1}{\beta_{33}^{(r)}} + \frac{1}{\gamma_{55}^{(r)}} + \frac{1}{\delta_{11}^{(r)}} \right) = 0,$$

where a term $(1/\gamma_{55}^{(4)})$ with value zero has been introduced so that each summation may run through the values $r = 1, 2, 3, 4$. In a similar way, by considering equilibrium at the joint, we may obtain the frequency equation for the second branch. It is

$$\frac{1}{\gamma_{44}^{(1)}} + \frac{1}{\beta_{44}^{(2)}} + \frac{1}{\gamma_{22}^{(2)}} + \frac{1}{\delta_{22}^{(2)}} = 0.$$

The frequency equations have been expressed in terms of receptances simply because these are more readily available than impedances (9). However, impedances may be used, and the corresponding equations are all obtained by replacing $1/\beta_{44}$, etc. by B_{44} , etc. Thus the equation replacing equation (4.1) will be

$$B_{44} + \Gamma_{22} + \Delta_{22} = 0.$$

Once the branch frequencies have been obtained, the corresponding branch modes may easily be found, using receptance analysis. Thus for the branch shown in Figure 3(b) the shapes of the various points of the branch will be given by the equations

$$q_{xb} = \beta_{x4} Q_{4b}, \quad q_{xc} = \gamma_{x2} Q_{2c}, \quad q_{xd} = \delta_{x2} Q_{2d},$$

where β_{x4} , for instance, is the receptance linking the displacement at position x with a couple at the end of the beam. The displacements may all be obtained in terms of the arbitrary quantity q_2 , thus

$$q_{xb} = \frac{\beta_{x4}}{\beta_{44}} q_2, \quad q_{xc} = \frac{\gamma_{x2}}{\gamma_{22}} q_2, \quad q_{xd} = \frac{\delta_{x2}}{\delta_{22}} q_2.$$

These equations define a mode shape for the complete branch. *However*, it will be shown in the following section that, unless the actual mode *shapes* of the whole system are required, it is unnecessary to find the shapes of the branch modes; all that is required is the branch *frequencies*.

5. THE RAYLEIGH-RITZ ANALYSIS

First consider an arbitrary continuous vibrating system for which expressions for the kinetic and potential energies T and V are known. The vibration of the system will be specified by the values of certain displacements at each point of the system. Suppose that the displacements to be specified at each point are $y_1, y_2, \dots y_s$; they will all be functions of the co-ordinates of the point. The displacements and co-ordinates may be taken to be the elements of column vectors $\mathbf{y}$ and $\mathbf{r}$ respectively. Suppose that the system may vibrate only in a combination of generalized modes $\mathbf{y}^{(1)}(\mathbf{r}), \mathbf{y}^{(2)}(\mathbf{r}), \dots \mathbf{y}^{(n)}(\mathbf{r})$. In harmonic motion with radian frequency ω the vector of generalized displacements corresponding to the point $\mathbf{r}$ of the system may be expressed in the form

$$\mathbf{y}(\mathbf{r}, t) = \sum_{r=1}^n u_r \mathbf{y}^{(r)}(\mathbf{r}) \sin \omega t.$$

For such a motion the kinetic and potential energies of the system will have the form

$$\begin{aligned} T &= \frac{1}{2}(\mathbf{u}' \mathbf{A} \mathbf{u}) \omega^2 \cos^2 \omega t \\ &= \frac{1}{2}(a_{11} u_1^2 + 2a_{12} u_1 u_2 + \dots + a_{nn} u_n^2) \omega^2 \cos^2 \omega t, \\ V &= \frac{1}{2}(\mathbf{u}' \mathbf{C} \mathbf{u}) \sin^2 \omega t \\ &= \frac{1}{2}(c_{11} u_1^2 + 2c_{12} u_1 u_2 + \dots + c_{nn} u_n^2) \sin^2 \omega t, \end{aligned}$$

where the inertia matrix $\mathbf{A}$ and stability matrix $\mathbf{C}$ have order n and are functions of the modes $\mathbf{y}^{(1)}(\mathbf{r}), \mathbf{y}^{(2)}(\mathbf{r}), \dots \mathbf{y}^{(n)}(\mathbf{r})$.

In the Rayleigh-Ritz method (11) we find the stationary values of the quotient

$$\omega_R^2 = \frac{\mathbf{u}' \mathbf{C} \mathbf{u}}{\mathbf{u}' \mathbf{A} \mathbf{u}}.$$

This leads to the eigenvalue equation

$$(\mathbf{C} - \omega^2 \mathbf{A}) \mathbf{u} = \mathbf{0} \quad (5.1)$$

for the appropriate natural frequencies ω and corresponding "mode shapes" $\mathbf{u}$. The vector $\mathbf{u}$ is not a "mode shape" in the usual sense. All that it gives is the proportions in which the vectors $\mathbf{y}^{(r)}(\mathbf{r})$ occur in the proper mode shape $\mathbf{y}$.

For a frame made up of beams in flexure the kinetic and potential energies will be

$$\begin{aligned} T &= \frac{1}{2} \sum \int A \rho y^2(x) dx, \\ V &= \frac{1}{2} \sum \int EI \left(\frac{\partial^2 y}{\partial x^2} \right)^2 dx, \end{aligned}$$

where the summation is over all the components and the integration is along a typical member. If the system is allowed to vibrate with frequency ω in a combination of modes, then the displacement at a typical point of any beam will take the form

$$y(x, t) = \left\{ \sum_{r=1}^n u_r y^{(r)}(x) \right\} \sin \omega t.$$

By substituting this expression for y in the equations for T and V we find that the elements of the matrices **A** and **C** are

$$a_{rs} = \sum \int A \rho y^{(r)}(x) y^{(s)}(x) dx,$$

$$c_{rs} = \sum \int EI y^{(r)}(x) y^{(s)}(x) dx,$$

where the summation and integration are as before.

We shall now obtain explicit expressions for these inertia and stability coefficients a_{rs} and c_{rs} . First, consider two modes *belonging to the same branch*. Since these modes are principal modes of the system when it is constrained in some way they will be *orthogonal*. Thus for two modes belonging to the same branch we have

$$a_{rs} = 0 = c_{rs},$$

and

$$c_{rr} = \omega_r^2 a_{rr}, \quad (5.2)$$

where ω_r is the natural frequency corresponding to that branch. Thus for modes belonging to any one branch it is sufficient to calculate the direct inertia coefficients a_{rr} . We shall now find these.

If a uniform beam is vibrating with frequency ω , its displacement $y(x)$ satisfies the equation

$$y''''(x) = (A \rho \omega^2 / EI) y = \lambda^4 y. \quad (5.3)$$

Rayleigh (13) shows that

$$4\lambda^4 \int_0^l y^2 dx = [3yy''' - y'y'' + x\{\lambda^4 y^2 - 2y'y''' + (y'')^2\}]_0^l,$$

so that

$$4\omega^2 \int_0^l A \rho y^2 dx = [3y(EI y''') - y'(EI y'') + x\{EI \lambda^4 y^2 - 2y'(EI y''') + (EI y'')^2 / EI\}]_0^l.$$

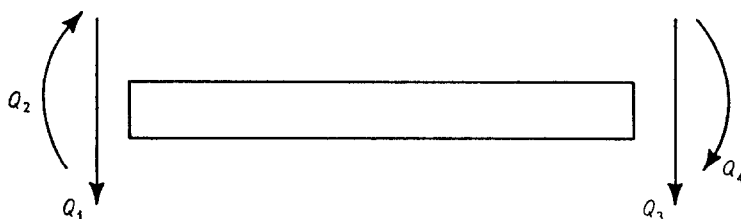


Figure 9. Shearing forces and bending moments at the ends of a beam in flexure.

Referring to Figure 9 we see that

$$y = q_1, \quad y' = q_2, \quad EI y'' = -Q_2, \quad EI y''' = Q_1, \quad \text{at } x = 0,$$

$$y = q_3, \quad y' = q_4, \quad EI y'' = Q_4, \quad EI y''' = -Q_3, \quad \text{at } x = l,$$

and thus

$$4\omega^2 \int_0^l A \rho y^2 dx = -3q_1 Q_1 - q_2 Q_2 - 3q_3 Q_3 - q_4 Q_4 + l\{EI \lambda^4 q_3^2 + 2q_4 Q_3 + Q_4^2 / EI\}. \quad (5.4)$$

In order to calculate the cross coefficients we use the equations

$$\omega_r^2 \int_0^l A\rho y^{(r)} y^{(s)} dx = [(EI y^{(r)''''}) y^{(s)} - (EI y^{(r)''}) y^{(s)''}]_0^l + \int_0^l EI y^{(r)''} y^{(s)''} dx,$$

$$\omega_s^2 \int_0^l A\rho y^{(r)} y^{(s)} dx = [y^{(r)} (EI y^{(s)''''}) - y^{(r)'} (EI y^{(s)''})]_0^l + \int_0^l EI y^{(r)''} y^{(s)''} dx.$$

These yield

$$\int_0^l A\rho y^{(r)} y^{(s)} dx = \frac{h_{rs} - h_{sr}}{\omega_r^2 - \omega_s^2}, \quad \int_0^l EI y^{(r)''} y^{(s)''} dx = \frac{\omega_r^2 h_{rs} - \omega_s^2 h_{sr}}{\omega_r^2 - \omega_s^2}, \quad (5.5)$$

where

$$h_{rs} = \sum_{i=1}^4 q_i^{(r)} Q_i^{(s)}, \quad (5.6)$$

the superscript over a quantity denoting its value in the mode concerned. The inertia and stability coefficients for the complete frame may now be obtained by applying equations (5.5) and (5.6) to each member.

It may be that, in addition to vibrating transversely, a beam vibrates as a rigid body. In this case extra terms have to be added to the inertia coefficients. The most common case is when the beam executes rigid-body vibrations in the direction of its length. Suppose that the rigid-body vibration is given by

$$w = \sum_{r=1}^n u_r w^{(r)}(x),$$

where now the "functions" $w^{(r)}(x)$ will be constant along the length of the beam. The kinetic energy due to this motion will be

$$T = \frac{1}{2} A\rho l \dot{w}^2,$$

so that the additions to the inertia coefficients will be

$$a'_{rr} = A\rho l (w^{(r)})^2, \quad a'_{rs} = A\rho l w^{(r)} w^{(s)}.$$

Put

$$w = q_5, \quad w^{(r)} = q_5^{(r)}.$$

Then the term which must be added to the right-hand side of equation (5.4), i.e. $8\omega^2 T$, will be

$$4\omega^2 A\rho l q_5^2.$$

The modification of the cross coefficients can be effected simply by extending the summation in equation (5.6) to include $i = 5$. For the amplitude of the longitudinal force Q_5 applied to the beam will be

$$Q_5 = -A\rho l \omega^2 q_5$$

so that a'_{rs} may be written

$$\begin{aligned} a'_{rs} &= \{q_s^{(r)}(-A\rho l\omega_s^2 q_s^{(s)}) - q_s^{(s)}(-A\rho l\omega_r^2 q_s^{(r)})\}/(\omega_r^2 - \omega_s^2), \\ &= (q_s^{(r)} Q_s^{(s)} - q_s^{(s)} Q_s^{(r)})/(\omega_r^2 - \omega_s^2). \end{aligned}$$

It may be verified that extending the summation in this way will leave c_{rs} unchanged. The direct coefficient c_{rr} for a branch will still be given by equation (5.2).

For numerical work it is convenient to employ dimensionless quantities. Suppose that the frame consists of a number of members $B, C, \dots U$ and that the dimensions of the typical member B are l_b , $(A\rho)_b$ and $(EI)_b$. Let l_0 , $(A\rho)_0$ and $(EI)_0$ be some standard dimensions and put

$$\bar{l}_b = l_b/l_0, \quad (\bar{A\rho})_b = (A\rho)_b/(A\rho)_0, \quad (\bar{EI})_b = (EI)_b/(EI)_0.$$

Similarly, introduce the dimensionless quantities given by

$$\bar{q}_{ib} = \begin{cases} q_{ib}/l_0 & , \\ q_{ib} & \end{cases}, \quad \bar{Q}_{ib} = \begin{cases} l_0^2 Q_{ib}/(EI)_0 & i \text{ odd}, \\ l_0 Q_{ib}/(EI)_0 & i \text{ even}. \end{cases}$$

Introduce a standard frequency ω_0 given by

$$\omega_0^2 = (EI)_0/(A\rho)_0 l_0^4 \quad (5.7)$$

and put

$$\bar{\omega} = \omega/\omega_0.$$

Finally put

$$\bar{a}_{rs} = a_{rs}/a_0, \quad \bar{c}_{rs} = c_{rs}/c_0,$$

where

$$a_0 = (EI)_0/l_0 \omega_0^2, \quad c_0 = (EI)_0/l_0.$$

Then the dimensionless equations for the inertia and stability coefficients of the whole structure are

$$\begin{aligned} 4\bar{\omega}_r^2 \bar{a}_{rr} &= \sum_b \{ -3\bar{q}_{1b} \bar{Q}_{1b} - \bar{q}_{2b} \bar{Q}_{2b} - 3\bar{q}_{3b} \bar{Q}_{3b} - \bar{q}_{4b} \bar{Q}_{4b} + \\ &\quad + l_b[(\bar{A\rho})_b \bar{\omega}_r^2 (\bar{q}_{3b}^2 + 4\bar{q}_{5b}^2) + 2\bar{q}_{4b} \bar{Q}_{3b} + \bar{Q}_{4b}^2/(\bar{EI})_b] \}, \end{aligned} \quad (5.8)$$

$$\bar{c}_{rr} = \bar{\omega}_r^2 \bar{a}_{rr}, \quad (5.9)$$

and

$$\bar{a}_{rs} = \frac{\bar{h}_{rs} - \bar{h}_{rs}}{\bar{\omega}_r^2 - \bar{\omega}_s^2}, \quad \bar{c}_{rs} = \frac{\bar{\omega}_r^2 \bar{h}_{rs} - \bar{\omega}_s^2 \bar{h}_{sr}}{\bar{\omega}_r^2 - \bar{\omega}_s^2}, \quad (5.10)$$

where

$$\bar{h}_{rs} = \sum_b \sum_{i=1}^5 \bar{q}_{ib}^{(r)} \bar{Q}_{ib}^{(s)}. \quad (5.11)$$

We have shown that, in order to find the inertia and stability coefficients, we need to know the displacements, slopes, forces and couples at the end of each beam. On inspection we find that there are only a very limited number of combinations of end conditions, which makes it desirable to tabulate them. The procedure used to tabulate these combinations can be seen by referring back to Figure 3(b). The beam B is clamped at its lower end and pinned at the upper one. Suppose that a couple Q_4 acts at the pinned end. Then the displacements and slopes at the ends will be

$$q_1 = 0 = q_2 = q_3, \quad q_4 = \beta_{44} Q_4 = \frac{F_3}{EI\lambda F_5} Q_4,$$

where the receptance has been taken from Table 3, line 3, of Bishop (7). The forces and couples acting at the ends may be found by relaxing one condition at a time. If, for instance, the beam is supposed to be *pinned* at the left-hand end, the couple Q_2 which must be applied there to make q_2 zero (as it is for a clamped end) will be given by

$$q_2 = 0 = \beta'_{22} Q_2 + \beta'_{24} Q_4,$$

where the receptances refer to a *pinned*-pinned beam.

Thus

$$Q_2 = -\frac{\beta'_{24}}{\beta'_{22}} Q_4 = \frac{F_8}{F_5} Q_4.$$

Similarly we find that

$$Q_1 = \frac{\lambda F_{10}}{F_5} Q_4, \quad Q_3 = \frac{\lambda F_1}{F_5} Q_4.$$

If we work in terms of dimensionless quantities and choose Q_4 to make $\bar{q}_4$ unity we obtain,

$$\{\bar{Q}_1, \bar{Q}_2, \bar{Q}_3, \bar{Q}_4, \bar{Q}_5\} = (\bar{EI})_b (\lambda_b l_0)^2 \left\{ \frac{F_{10}}{F_3}, \frac{F_8}{\lambda_b l_0 F_3}, \frac{F_1}{F_3}, \frac{F_5}{\lambda_b l_0 F_3}, 0 \right\}.$$

It should be remembered that if the beams have different dimensions the values of " λ " for the various beams will be different; hence the use of the notation λ_b . The other possible cases have been tabulated in Table 1. In this table ν , λl and F_s are used for $\lambda_b l_0$, $\lambda_b l_b$ and $F_s(\lambda_b l_b)$ respectively. Entry 5 has been included because rigid-body motions sometimes have to be considered, as in Figure 5. Entries 1-5 are the basic ones since entries 6-13 may all be obtained by taking suitable combinations of the first four. For beams having neither end clamped (entries 6-13) there have to be two entries because the beam may be excited at each end.

It was stated in section 2 that Table 1 may be used to give the values of impedances. Lines 1-5 of this table give the forces and couples which must be applied at the ends of a beam when one of the co-ordinates is allowed to vary independently of the others. Lines 6-13 give certain combinations of forces and couples which will produce certain combinations of deflections and slopes. Let us apply the results shown in the table to the system of Figure 2(a). For the beam *D* the only non-zero co-ordinates are q_2 and q_4 , and these are independent. Thus the impedances Δ_{22} and Δ_{24} should appear in line 2, columns 2 and 4, and the impedances Δ_{42} and Δ_{44} in positions 42 and 44; this is seen to be true. The beam *C* is described by line 8— q_3 is not zero, and in fact depends on q_2 —and the impedance Γ_{22} is indeed found in position 2. It is believed that by using this table in this way all ambiguity in the idea of an impedance may be removed.

6. MATRIX ANALYSIS

The analysis of a complicated frame may conveniently be carried out in matrix form. It has been shown that the inertia and stability matrices for the Rayleigh-Ritz analysis may be obtained from the values of the displacement, slopes, shearing forces and bending moments at the ends of the members of the frame. In this section it will be assumed that these quantities have been made dimensionless, as shown in the previous section, and the bars over these quantities will therefore be omitted.

Consider a frame made up of m beams, $B, C \dots U$ for which a total of n branch modes

is to be used. For the typical beam B vibrating in the r th mode we may form the column matrices

$$\mathbf{x}_b^{(r)} = \{q_{1b}^{(r)}, q_{2b}^{(r)}, q_{3b}^{(r)}, q_{4b}^{(r)}, q_{5b}^{(r)}\},$$

$$\mathbf{f}_b^{(r)} = \{Q_{1b}^{(r)}, Q_{2b}^{(r)}, Q_{3b}^{(r)}, Q_{4b}^{(r)}, Q_{5b}^{(r)}\}.$$

The various vectors corresponding to the different beams may be placed end to end to form two column matrices

$$\mathbf{x}^{(r)} = \{\mathbf{x}_b^{(r)} | \mathbf{x}_c^{(r)} | \dots | \mathbf{x}_u^{(r)}\}, \quad (6.1)$$

$$\mathbf{f}^{(r)} = \{\mathbf{f}_b^{(r)} | \mathbf{f}_c^{(r)} | \dots | \mathbf{f}_u^{(r)}\}, \quad (6.2)$$

of orders $5m$, for each mode. Finally the various column matrices corresponding to the different modes may be placed side by side to form two $5m \times n$ matrices

$$\mathbf{X} = [\mathbf{x}^{(1)} | \mathbf{x}^{(2)} | \dots | \mathbf{x}^{(n)}], \quad (6.3)$$

$$\mathbf{F} = [\mathbf{f}^{(1)} | \mathbf{f}^{(2)} | \dots | \mathbf{f}^{(n)}]. \quad (6.4)$$

We shall now express equations (5.8) to (5.10) in terms of the matrices $\mathbf{X}$ and $\mathbf{F}$. First we find the "direct" inertia coefficients. Equation (5.8) (in "unbarred" form) may be written.

$$a_{rr} = \mathbf{x}'^{(r)} \mathbf{L} \mathbf{x}^{(r)} + (1/\omega_r^2) \mathbf{x}'^{(r)} \mathbf{M} \mathbf{f}^{(r)} + (1/\omega_r^2) \mathbf{f}'^{(r)} \mathbf{N} \mathbf{f}^{(r)}, \quad (6.5)$$

where $\mathbf{L}$, $\mathbf{M}$, $\mathbf{N}$ are given by

$$\mathbf{L} = \text{diag}(\mathbf{L}_b, \mathbf{L}_c, \dots, \mathbf{L}_u),$$

$$\mathbf{M} = \text{diag}(\mathbf{M}_b, \mathbf{M}_c, \dots, \mathbf{M}_u),$$

$$\mathbf{N} = \text{diag}(\mathbf{N}_b, \mathbf{N}_c, \dots, \mathbf{N}_u),$$

in which

$$\mathbf{L}_b = \text{diag}(0, 0, l_b(A\rho)_b/4, 0, l_b(A\rho)_b),$$

$$M_b = \frac{1}{4} \begin{bmatrix} -3 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 & 0 \\ 0 & 0 & 2l_b & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

and

$$N_b = \text{diag}(0, 0, 0, l_b/4(EI)_b, 0).$$

(The matrices $\mathbf{L}$ and $\mathbf{N}$ are diagonal matrices, while $\mathbf{M}$ is a partitioned matrix with $\mathbf{M}_b, \mathbf{M}_c, \dots, \mathbf{M}_u$ in the principal diagonal blocks and zeros elsewhere.) Now put

$$\mathbf{W} = \text{diag}(\omega_1^2, \omega_2^2, \dots, \omega_n^2).$$

Then the right-hand side of equation (6.5) can be recognized as the r th diagonal element of the matrix

$$\mathbf{A}^* = \mathbf{X}' \mathbf{L} \mathbf{X} + \mathbf{W}^{-1} \mathbf{X}' \mathbf{M} \mathbf{F} + \mathbf{W}^{-1} \mathbf{F}' \mathbf{N} \mathbf{F}. \quad (6.6)$$

Thus the diagonal elements of $\mathbf{A}$ and $\mathbf{C}$ may be calculated as the diagonal elements of the matrices $\mathbf{A}^*$ and $\mathbf{C}^*$, where

$$\mathbf{C}^* = \mathbf{W} \mathbf{A}^*.$$

The cross coefficients may be obtained in matrix form by noticing that h_{rs} , given by equation (5.11), is the typical element of the matrix

$$\mathbf{H} = \mathbf{X}'\mathbf{F}. \quad (6.7)$$

For each value of r form the diagonal matrix

$$\mathbf{U}_r = \text{diag}(u_{r1}, u_{r2}, \dots, u_{rn}),$$

where

$$u_{rs} = \begin{cases} 1/(\omega_r^2 - \omega_s^2) & r \neq s \\ 0 & r = s. \end{cases}$$

Now form the matrix products

$$\mathbf{A}_r^{**} = \mathbf{U}_r(\mathbf{H}' - \mathbf{H}),$$

$$\mathbf{C}_r^{**} = \mathbf{U}_r\{(\mathbf{WH})' - (\mathbf{WH})\}.$$

The elements in the first column of $\mathbf{A}_1^{**}$ are

$$0, \frac{h_{12} - h_{21}}{\omega_1^2 - \omega_2^2}, \frac{h_{13} - h_{31}}{\omega_1^2 - \omega_3^2}, \dots, \frac{h_{1n} - h_{n1}}{\omega_1^2 - \omega_n^2},$$

and equation (5.10) shows that this first column is

$$0, a_{12}, a_{13}, \dots, a_{1n}.$$

Apart from the zero in the first position, this is the first column of the inertia matrix $\mathbf{A}$.

The general result is that the r th columns of $\mathbf{A}_r^{**}$ and $\mathbf{C}_r^{**}$ are the r th columns of $\mathbf{A}$ and $\mathbf{C}$ respectively, except that they have zeros in the r th place. Thus the matrix $\mathbf{A}$ (or $\mathbf{C}$) may be found by taking the first column of $\mathbf{A}_1^{**}$ (or $\mathbf{C}_1^{**}$), the second column of $\mathbf{A}_2^{**}$ (or $\mathbf{C}_2^{**}$) etc., and then adding in the diagonal terms which have already been found. This procedure is very easy to programme for a computer, provided that the matrix scheme which is being used has a facility for address modification, as in the Automat used with the Ferranti Pegasus.

When the matrices $\mathbf{A}$ and $\mathbf{C}$ have been found, the problem is reduced to finding the eigenvalues and eigenvectors of the matrix equation (5.1), namely

$$(\mathbf{C} - \omega^2 \mathbf{A})\mathbf{u} = \mathbf{0}. \quad (6.8)$$

This equation is best solved (11) by reducing it to an ordinary eigenvalue problem of the form

$$(\mathbf{B} - \omega^2 \mathbf{I})\mathbf{u} = \mathbf{0}.$$

This may be effected by factoring $\mathbf{A}$ into symmetric factors

$$\mathbf{A} = \mathbf{K}\mathbf{K}',$$

where $\mathbf{K}$ is a lower triangular matrix, and then introducing a new vector

$$\mathbf{v} = \mathbf{K}'\mathbf{u}.$$

Then

$$(\mathbf{C} - \omega^2 \mathbf{A})\mathbf{u} = (\mathbf{C} - \omega^2 \mathbf{K}\mathbf{K}')\mathbf{u} = (\mathbf{C}(\mathbf{K}^{-1})' - \omega^2 \mathbf{K})\mathbf{v} = \mathbf{0},$$

or

$$(\mathbf{K}^{-1} \mathbf{C} (\mathbf{K}^{-1})' - \omega^2 \mathbf{I})\mathbf{v} = \mathbf{0}.$$

This ordinary eigenvalue problem for the symmetric matrix

$$\mathbf{B} = \mathbf{K}^{-1} \mathbf{C}(\mathbf{K}^{-1})'$$

may now be solved using a standard programme for the eigenvalues and eigenvectors of a real symmetric matrix.

Suppose now that a number of natural frequencies and corresponding eigenvectors $\mathbf{u}^{(1)}, \mathbf{u}^{(2)}, \dots, \mathbf{u}^{(t)}$ have been found from equation (6.8). The elements $u_1^{(s)}, u_2^{(s)}, \dots, u_n^{(s)}$ of any one of these vectors give the proportions in which the n branch modes occur in the s th approximate principal mode of the system. This means that if the displacement, slope, shearing force or bending moment at a given point of a given beam, in the r th branch mode, is $z^{(r)}(x)$, then the value of the corresponding quantity in the s th approximate principal mode will be

$$\begin{aligned} w^{(s)}(x) &= u_1^{(s)} z^{(1)}(x) + u_2^{(s)} z^{(2)}(x) + \dots + u_n^{(s)} z^{(n)}(x) \\ &= [z^{(1)}(x), z^{(2)}(x), \dots, z^{(n)}(x)] \mathbf{u}^{(s)}. \end{aligned}$$

Putting together the equations for the displacement and slopes we find that the vector of q 's for the s th principal mode, corresponding to $\mathbf{x}^{(r)}$ of equation (6.1), is

$$\mathbf{y}^{(s)} = [\mathbf{x}^{(1)} | \mathbf{x}^{(2)} | \dots | \mathbf{x}^{(n)}] \mathbf{u}^{(s)} = \mathbf{X} \mathbf{u}^{(s)}.$$

Thus, if

$$\mathbf{Y} = [\mathbf{y}^{(1)} | \mathbf{y}^{(2)} | \dots | \mathbf{y}^{(t)}],$$

$$\mathbf{U} = [\mathbf{u}^{(1)} | \mathbf{u}^{(2)} | \dots | \mathbf{u}^{(t)}],$$

then

$$\mathbf{Y} = \mathbf{X} \mathbf{U}. \quad (6.9)$$

Similarly, the matrix $\mathbf{G}$ for the principal modes, corresponding to $\mathbf{F}$ for the branch modes, will be

$$\mathbf{G} = \mathbf{F} \mathbf{U}. \quad (6.10)$$

7. A SIMPLE APPLICATION

In this section we shall apply the Branch Mode method to the system shown in Figure 2(a). We shall give the analysis in detail for the symmetric modes and then give a brief discussion of the analysis leading to the anti-symmetric modes.

To find the symmetric modes we use the branches shown in Figure 3. For simplicity we shall consider the case in which B, C, D and E all have identical values of $l, A\rho$ and EI , which will be taken to be the standard values. In this case we shall have

$$l_b = l = (\overline{A\rho})_b = (\overline{EI})_b, \quad \lambda_b = \lambda, \quad \nu^4 = (\lambda l)^4 = \bar{\omega}^2,$$

for each of the beams, the last result being deduced from equations (5.3) and (5.7).

The equations governing the first branch are equations (4.1), namely

$$q_{4b} = \beta_{44} Q_{4b}, \quad q_{2c} = \gamma_{22} Q_{2c}, \quad q_{2d} = \delta_{22} Q_{2d},$$

and the compatibility and equilibrium relations

$$q_{4b} = q_{2c} = q_{2d},$$

$$Q_{4b} + Q_{2c} + Q_{2d} = 0.$$

In general, the condition that these equations should have a non-trivial solution is that equation (4.2) should hold. However, when B and D are identical, equation (4.2) gives

only the modes which are anti-symmetrical about C . The symmetrical modes are specified by

$$q_{4b} = 0 = q_{2c} = q_{2d},$$

$$Q_{4b} + Q_{2d} = 0, \quad Q_{2c} = 0. \quad (7.1)$$

In these modes, which are symmetrical about C , B and D each vibrate as clamped-clamped beams having the same amplitude and phase, and C does not vibrate at all. The frequencies corresponding to these modes are the natural frequencies of a clamped-clamped beam. Denoting the two sub-branches of the first branch by $1'$, $1''$ we find that there are three branches $1'$, $1''$ and 2 with frequency equations

$$F_5/F_3 + F_2/F_6 = 0; \quad F_3 = 0; \quad F_5/F_3 + 2F_2/F_6 = 0; \quad (7.2)$$

respectively. These equations may be solved very easily and accurately, using a simple iterative method. The first few zeros are found to be:

$$\left. \begin{array}{rcccl} 2.146 & 860 & 4 & & 2.029 & 519 & 6 \\ 4.103 & 476 & 3 & 4.730 & 040 & 8 & 4.197 & 319 & 7 \\ \hline 5.329 & 861 & 7 & & 5.239 & 069 & 6 \\ 7.238 & 208 & 2 & 7.853 & 204 & 6 & 7.330 & 004 & 0 \\ \hline 8.469 & 375 & 4 & & 8.377 & 447 & 3 \end{array} \right\} \quad (7.3)$$

The next step is to find the direct inertia coefficients a_{rr} . These may be found using the method described in section 6, but in the present simple problem may be found more easily by using equation (5.8) directly. The values of the q 's and Q 's to be used may be taken from Table 1. For the sub-branch $1'$ of that shown in Figure 3(b), the values for B , C and D are taken from lines 4, 8 and 2 respectively. We obtain

$$4\nu^4 a_{rr} = \left(\frac{-\nu F_5}{F_3} + \frac{2\nu^2 F_1}{F_3} + \frac{\nu^2 F_5^2}{F_3^2} \right) + \left(\frac{-2\nu F_2}{F_6} + \frac{\nu^2 F_{10}^2}{F_6^2} + \frac{\nu^2 F_9^2}{F_6^2} \right) + \left(\frac{-\nu F_5}{F_3} + \frac{\nu^2 F_8^2}{F_3^2} \right),$$

where ν has the value corresponding to the r th mode of the sub-branch $1'$. Using the fact that ν satisfies the first of equations (7.2) we find that

$$a_{rr} = \frac{1}{4\nu^2} \left(\frac{2F_1}{F_3} + \frac{F_5^2 + F_8^2}{F_3^2} + \frac{F_9^2 + F_{10}^2}{F_6^2} \right).$$

Using the same procedure for the branch 2 we find that

$$a_{rr} = \frac{1}{4\nu^2} \left(\frac{2F_1}{F_3} + \frac{F_5^2}{F_3^2} + \frac{F_9^2 + F_{10}^2}{F_6^2} \right).$$

For the sub-branch $1''$ the vector $\mathbf{x}$ of q 's [see equation (6.1)] is identically zero and the vector $\mathbf{f}$ may be taken to be

$$\mathbf{f} = \{\nu^2 F_{10}, \nu F_8, \nu^2 F_1, \nu F_5, 0 | 0 | \nu^2 F_1, -\nu F_5, \nu^2 F_{10}, \nu F_8, 0 | 0\}. \quad (7.4)$$

This vector has been formed from lines 4 and 2 of Table 1. The entries in these lines have been multiplied up by F_3 to make q_{b4} and q_{d2} zero, and the entry in line 2 has been multiplied by (-1) to satisfy the equilibrium relation (7.1). The two zero sub-vectors refer to the beams C and D . For this branch, equation (5.8) now gives

$$a_{rr} = (F_5^2 + F_8^2)/4\nu^2.$$

The values of the coefficients c_{rr} may be obtained from equation (5.2), namely

$$c_{rr} = \nu^4 a_{rr}.$$

The cross terms may be found by using the method described in the previous section. The vectors $\mathbf{x}^{(r)}$ and $\mathbf{f}^{(r)}$ will have order 20. The elements of $\mathbf{x}_b^{(r)}$, $\mathbf{x}_c^{(r)}$ and $\mathbf{x}_d^{(r)}$ in the sub-branch 1' may be obtained from lines 4, 8 and 2, respectively, of Table 1. In fact, if the column vectors of the q_r 's and Q_r 's in line s are denoted by $\mathbf{y}^{(s)}$ and $\mathbf{g}^{(s)}$ respectively, then in a mode of the first branch we have

$$\mathbf{x} = \{\mathbf{y}^{(4)}|\mathbf{y}^{(8)}|\mathbf{y}^{(2)}|\mathbf{o}\}; \quad \mathbf{f} = \{\mathbf{g}^{(4)}|\mathbf{g}^{(8)}|\mathbf{g}^{(2)}|\mathbf{o}\}.$$

In the sub-branch 1'', $\mathbf{x}$ is zero and $\mathbf{f}$ is given by equation (7.4), which will be written

$$\mathbf{x} = \mathbf{o}; \quad \mathbf{f} = \{\mathbf{g}^{*(4)}|\mathbf{o} - \mathbf{g}^{*(2)}|\mathbf{o}\}.$$

In branch 2 we have

$$\mathbf{x} = \{\mathbf{o}|\mathbf{o}|\mathbf{y}^{(4)}|\mathbf{y}^{(8)}\}; \quad \mathbf{f} = \{\mathbf{o}|\mathbf{o}|\mathbf{g}^{(4)}|\mathbf{g}^{(8)}\}.$$

These modes, evaluated for the appropriate branch frequencies, can be used for the columns of the matrices $\mathbf{X}$ and $\mathbf{F}$, according to equations (6.3) and (6.4). However, since the direct coefficients have already been calculated, simpler vectors may be used. For it is known that the only non-zero cross terms are those corresponding to two modes belonging to different branches. Thus the only terms which need to be calculated in an $\mathbf{x}$ (or $\mathbf{f}$) belonging to one branch are those which are multiplied by a non-zero factor occurring in an $\mathbf{f}$ (or $\mathbf{x}$) of another branch. If this criterion is used to eliminate terms it is found that the following vectors may be used:

$$\begin{aligned} \mathbf{x} &= \{\mathbf{o}|\mathbf{o}|\mathbf{o}, \mathbf{1}, \mathbf{o}, \mathbf{o}, \mathbf{o}|\mathbf{o}\}, & \mathbf{f} &= \{\mathbf{o}|\mathbf{o}|\mathbf{o}, \mathbf{o}, \mathbf{o}, \nu F_8/F_3, \mathbf{o}|\mathbf{o}\}, \\ \mathbf{x} &= \{\mathbf{o}|\mathbf{o}|\mathbf{o}|\mathbf{o}\}, & \mathbf{f} &= \{\mathbf{o}|\mathbf{o}|\mathbf{o}, \mathbf{o}, \mathbf{o}, -\nu F_8, \mathbf{o}|\mathbf{o}\}, \\ \mathbf{x} &= \{\mathbf{o}|\mathbf{o}|\mathbf{o}, \mathbf{o}, \mathbf{o}, \mathbf{1}, \mathbf{o}|\mathbf{o}\}, & \mathbf{f} &= \{\mathbf{o}|\mathbf{o}|\mathbf{o}, \nu F_8/F_3, \mathbf{o}, \mathbf{o}, \mathbf{o}|\mathbf{o}\}. \end{aligned}$$

The procedure described above was carried out using, successively, 5, 7, 10 and 12 branch modes as indicated by the straight rules in (7.3). The results are shown in Table 2 where the straight rules indicate the position in the frequency scale of the highest branch frequency that was taken into account. The "exact" frequencies shown in Table 2 were calculated from the impedance equation (2.4). It is perhaps worth recording that, even in this simple problem, much more trouble was encountered in computing the zeros of the impedance determinant than in computing the eigenvalues from the Branch Mode analysis. Because of the arbitrary behaviour of the determinant, treated as a function of ω , it is very easy to miss a zero altogether.

Inspection of Table 2 shows that the method gives acceptable approximations to those natural frequencies which are below the upper limit of the branch frequencies. For the higher frequencies of the two-storey frame, longitudinal vibrations should be taken into account, and these will modify the "exact" values shown. However, the primary purpose of the analysis has been to compare the "Impedance Determinant" and Branch Mode methods, and in this comparison longitudinal motions have deliberately been omitted. They may easily be included in both methods.

Table 2 clearly shows the convergence of the approximate natural frequencies to be the exact ones. Nevertheless, it is perhaps of some interest to show, independently, that the aggregate of the modes of the two branches which have been used do indeed form a complete set for the symmetrical vibrations of the frame. To prove this we use the fact that the set of principal modes of a system forms a complete set for that system. In other

words, any (well-behaved) displacement function satisfying given end-conditions may be expressed as a linear combination of (the infinity of) principal modes of the system corresponding to these end-conditions. To prove the final result for the two-storey frame under consideration we notice that any principal mode of the (half-) frame may be expressed as the superposition of two parts, one satisfying the end conditions of the first, and the other satisfying the end conditions of the second branch. These two parts may be expressed as linear combinations of the principal modes of the first and second branches respectively—in virtue of the result just quoted—that is, of the branch modes. Therefore, any principal mode of the whole system may be expressed in terms of the branch modes. The matter may perhaps be made clearer if attention is focused on the vector of end

TABLE 2
Error analysis for branch mode approximations to symmetrical modes

	Exact Frequencies	5 Branch Modes	% Error	7 Branch Modes	% Error	10 Branch Modes	% Error	12 Branch Modes	% Error
1	3·9124	3·9605	1·2	3·9528	1·0	3·9204	0·21	3·9134	0·025
2	4·6866	4·6911	0·10	4·6887	0·04	4·6870	0·008	4·6867	0·001
3	14·8186	15·4105	4·0	15·3025	3·3	14·9061	0·59	14·8287	0·068
4	20·3441	20·6728	1·6	20·4910	0·72	20·3657	0·11	20·3465	0·012
5	26·6516	33·5543	26	26·7662	0·43	26·6683	0·062	26·6533	0·006
6	28·6143			28·7952	0·63	28·6354	0·074	28·6164	0·007
7	48·7989			68·1559	40	48·8248	0·053	48·8027	0·008
8	58·3556					58·5635	0·36	58·3632	0·013
9	68·9144					82·7845	20	68·9248	0·015
10	72·0594					174·053	142	72·0671	0·011

displacement and slopes, rather than on the actual mode shapes. For any principal mode of the half-frame the vector $\mathbf{x}$ has the form

$$\mathbf{x} = \{q_2 \mathbf{z}^{(4)} | q_2 \mathbf{z}^{(8)} | q_2 \mathbf{z}^{(2)} + q_4 \mathbf{z}^{(4)} | q_4 \mathbf{z}^{(8)}\},$$

where the $\mathbf{z}$'s are evaluated for the appropriate frequency parameter and the ratio $q_2:q_4$ is given by equations (2.4). The vector $\mathbf{x}$ may be expressed as the sum of two parts, namely

$$\mathbf{x} = q_2 \{\mathbf{z}^{(4)} | \mathbf{z}^{(8)} | \mathbf{z}^{(2)} | \mathbf{0}\} + q_4 \{\mathbf{0} | \mathbf{0} | \mathbf{z}^{(4)} | \mathbf{z}^{(8)}\}.$$

The first of these vectors satisfies the end conditions of the first branch, and may therefore be expressed as a linear combination of the totality of $\mathbf{x}$ -vectors corresponding to the modes in the first branch (sub-branches 1' and 1"). The second satisfies the end conditions of the second branch and may therefore be expressed as a linear combination of the $\mathbf{x}$ -vectors of the modes in the second branch. It is clear that the foregoing argument may be generalized to show the completeness of the set of branch modes, chosen as in section 3, for a general frame.

Exact and approximate mode shapes for the first six principal modes are shown in Figures 10 and 11. It will be seen that, with five branch modes taken into account, the method gives shapes for the first two modes that are exact to within the accuracy which can be represented in the diagrams, and shapes for the third and fourth modes which are still quite accurate. Figure 11 shows that to obtain the fifth and sixth mode shapes it is sufficient to take seven branch modes into account.

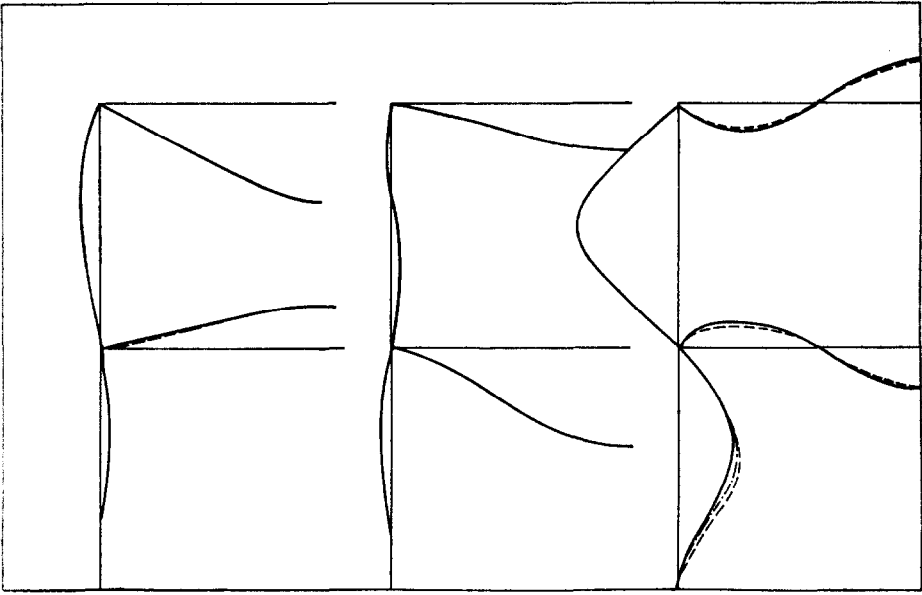


Figure 10. Exact and approximate shapes for the first three symmetrical modes.

— exact mode shape; - - - approximation using 5 branch modes; - · - · - approximation using 7 branch modes.

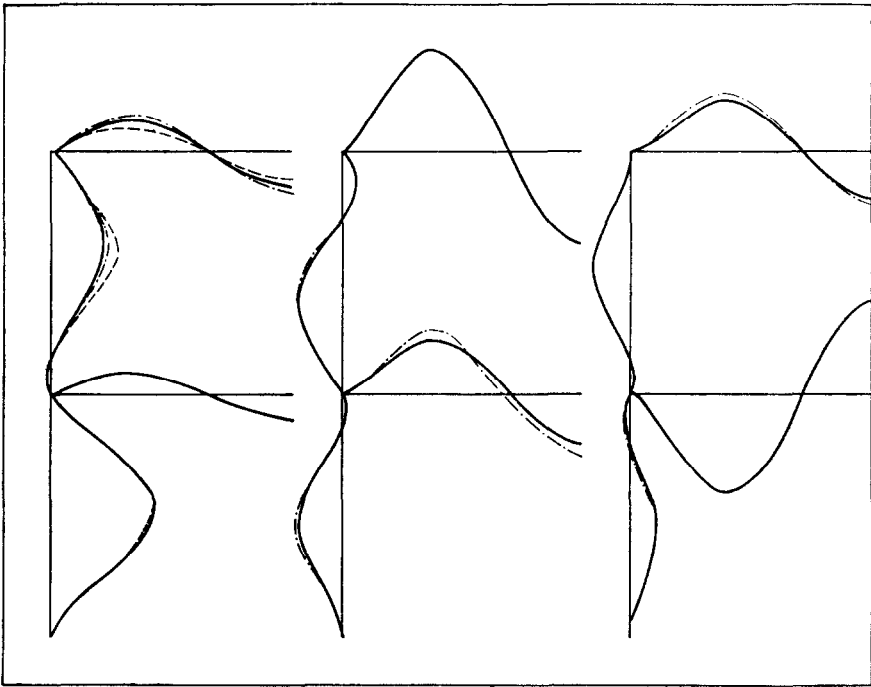


Figure 11. Exact and approximate shapes for the fourth, fifth and sixth symmetrical modes.

— exact mode shape; - - - approximation using 5 branch modes; - · - · - approximation using 7 branch modes.

The anti-symmetrical vibrations of the two-storey frame were analysed in the same way, using the four branches shown in Figure 5. The frequency equations of the branches were found to be

$$2F_6/F_3 + \nu = 0; \quad F_5/F_3 - F_1/F_5 = 0; \quad F_6/F_3 + \nu = 0; \quad F_5/F_3 - 2F_1/F_5 = 0;$$

respectively. The first few branch frequencies were found to be

1.923 535 7	3.496 511 7	1.718 881 3	3.393 231 4
4.999 953 0	4.358 054 0	4.892 769 9	4.463 324 2
8.049 920 4	6.648 462 2	7.964 460 2	6.545 413 8

The approximate natural frequencies obtained by taking successively 4, 8, 12 of these branch modes are shown in Table 3. It will be seen that, when only four branch modes are taken into account, the percentage errors in the fundamental and first overtone

TABLE 3
Error analysis for branch mode of approximations to antisymmetrical modes

	Exact Frequencies	4 Branch Modes	% Error	8 Branch Modes	% Error	12 Branch Modes	% Error
1	1.0554	1.4373	36	1.0627	0.69	1.0560	0.054
2	3.7165	4.1113	11	3.7278	0.30	3.7173	0.021
3	10.8262	11.0241	1.8	10.8313	0.047	10.8269	0.006
4	12.9819	<u>13.3060</u>	<u>2.5</u>	12.9850	0.024	12.9820	0.001
5	18.8256			18.8426	0.090	18.8282	0.014
6	24.1133			<u>24.1298</u>	<u>0.068</u>	24.1161	0.012
7	41.2266			54.2857	32	41.2303	0.009
8	45.5200			71.2981	57	45.5257	0.013
9	55.4848					<u>55.8081</u>	<u>0.58</u>

frequencies are large—larger than would be obtained using some other approximate methods—but that when more branch modes are considered a large number of the natural frequencies are obtained almost exactly. The corresponding mode shapes have not been shown because it is very difficult to obtain “exact” mode shapes (from the impedance method) with which the approximate mode shapes can be compared. Thus, even in this simple problem, it is easier to obtain the mode shapes from the Branch Mode method than from the impedance analysis.

8. CONCLUSIONS

The method presented in this paper was devised primarily because of the practical difficulty of obtaining natural frequencies, and particularly mode shapes, of complicated frame systems from receptance or impedance analysis. For the simple system to which it is applied in the paper the method is shown to give very accurate results, and it is hoped to develop it so that it can be applied to a more realistic multi-storey space frame. The method has the advantage of being essentially simple, and of leading to an eigenvalue problem, which is particularly suitable for computational work.

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